Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/20/2004
PI Name:	Mansur, Louis K. Ph.D.		
Project Title:	Materials Science and Particle Transport Assessment of Novel and Multifunctional Shielding Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	37831-6138	**Congressional District:**	3
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Flight Program:			
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Task Description:	Our approach consists of 1.) Specially preparing advanced shielding materials, 2.) Characterizing their structure and properties, 3.) Carrying out GCR transport and beam transport calculations for both pre-experiment planning and post-measurement analyses, and 4.) Measuring particle types and energy spectra during beam irradiation experiments at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. In-depth assessments will be carried out with respect to physical, mechanical and shielding properties. Properties such as structural strength, hydrogen storage, potential for thermal management, and electrical attributes will be assessed on a material specific basis. Calculations of shielding effectiveness will be accompanied by high energy accelerator experiments at the NASA Space Radiation Laboratory. A primary aim of these experiments will be to determine shielding effectiveness. An equally important goal will be to measure the response of the materials to environmental variables. For example, the extent of loss of hydrogen or of changes in physical or mechanical properties will be important. The effect of other types of loads will be evaluated. For example, the effects of aging, of temperature and of temperature cycling on properties will be assessed. The purpose of this phase of evaluations is to establish whether a promising material can maintain its desirable attributes. There is a high priority need to evaluate new materials for spacecraft shielding in deep space missions. This research will carry out new work on space radiation shielding materials. Both novel shielding materials and multifunctional spacecraft materials that incorporate desirable shielding properties will be investigated. The aim is to clearly establish the most promising materials and to assess their key properties with respect to space radiation shielding and suitability for other spacecraft applications. The approach will emphasize materials science. The required nuclear properties, especially the overriding need to incorporate the highest possible proportion of light nuclei, especially hydrogen, will be kept in constant view. The identification, development and qualification of new and multifunctional shielding materials are of the high importance to future manned NASA space missions. Knowledgeable researchers have even gone further to state that new shielding materials must be developed in order to enable long term deep space missions. Based on new understanding it is now clear that only those shielding materials that contain high proportions of hydrogen and other light elements, and with an absolute minimum of heavy elements, can be effective against galactic cosmic rays. It is a disturbing realization that recent transport calculations for galactic cosmic rays (GCR), coupled with currently proposed calculational models of biological effects have led to an understanding that conventional spacecraft materials like aluminum alloys may result in radiation effects that are more detrimental than would result with no shielding at all. This effect occurs through fragmentation of the incoming cosmic rays and the heavy nuclei of the spacecraft material such that the sum of the biological damage from the resulting new charged particles and neutrons exceeds that of the original cosmic ray particle. Conventional models of biological effect are based on risk assessment dictated by the magnitude of linear energy transfer (LET) of a particular particle at a given energy, coupled with accepted quality factors that calibrate LET in terms of relative biological effectiveness (RBE) (ICRP, 1991). It can be characterized as a dose equivalent method. While this approach may often be useful as a first approximation, it cannot recognize particular mechanisms of action of radiation on organisms. Heavy charged particles may interact with cellular molecules and subassemblies in many different ways, producing detrimental effects that are physically inaccessible to X-rays no matter how high the X-ray dose. An example based on radiation damage physics even without considering a detailed biological pathway is that charged particles with energies as low as a few keV may displace atoms from their molecular sites or cellular subassembly sites to locations far removed, in addition to producing ionization and excitation. X-rays on the other hand cannot generally displace atoms. They produce ionization and excitation. [For a photon to produce displacements to a measurable degree, it generally must be in the energy range of gamma rays above 1.5 MeV (Mansur and Farrell, 1997). While the gamma ray does not produce displacements directly, the resulting Compton electrons and electron/positron pairs can have enough energy to displace atoms.] Thus, there are compelling physical reasons on the most basic level that heavy charged particles may have different and more severe effects than would be obtained by any multiple of X-ray dose.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Radiation doses from galactic cosmic rays (GCR) are a significant issue for spacecraft crew exposures in deep space, which could severely constrain the duration of missions. We have completed and reported our initial work to evaluate a range of possible materials for GCR shielding. Earlier work had shown that conventional spacecraft structural materials, such as aluminum and higher atomic number structural alloys, provide relatively little shielding and, under certain conditions, may increase radiation risk. Materials containing high proportions of hydrogen and other low atomic mass number nuclei provide improved GCR shielding. Polyethylene (PE) is generally considered a good performance benchmark shield material. However, PE shielding occupies volume and adds mass to the spacecraft. In this work we have investigated several materials that are shown to provide shielding similar to PE, but which could furnish additional spacecraft functions such as structural support and thermal management, for example, possibly eliminating the need for materials currently used for these purposes. Carbon forms that can incorporate a large mass of hydrogen, as well as polymers and polymer composites are being explored. Calculations of shielding effectiveness in GCR spectra have been carried out. Experiments to measure shielding metrics recently have been completed at the NASA Space Radiation Laboratory (NSRL) located at Brookhaven National Laboratory (BNL) using beams of 1 GeV/nucleon and 0.6 GeV/nucleon O16. In a peer reviewed paper we have reported preliminary results for PE-infiltrated carbon forms (foams and fiber monoliths) and for several polymer materials.
Bibliography Type:	Description: (Last Updated: 08/08/2007)
Articles in Peer-reviewed Journals	L. K. Mansur, B. J. Frame, N. C. Gallego, S. B. Guetersloh, J. O. Johnson, J. W. Klett, and L. W. Townsend, "Assessment of Shielding Material Performance for Deep Space Missions," Proceedings of the Materials Research Society, Symposium NN on Materials for Space Applications, Volume 851, Materials Research Society, Warrendale, PA , Mar-2005
Presentation	L. K. Mansur,T. D. Burchell, B. J. Frame, N. C. Gallego,C. J. Janke, J. O. Johnson, L. W. Townsend "Materials Science and Particle Transport Assessment of Novel and Multifunctional Shielding Materials, " Third International Workshop on Space Radiation Research, Port Jefferson, New York, May 18, 2004 May-2004

Presentation	L. K. Mansur, T. D. Burchell, B. J. Frame, N. C. Gallego,C. J. Janke, J. O. Johnson, L. W. Townsend, "Progress in Materials Science and Particle Transport Assessment of Novel and Multifunctional GCR Shielding Materials," Space Radiation Shielding Principal Investigators Meeting, National Space Science and Technology Center, Huntsville, Alabama, August 19-20, 2004 Aug-2004
Presentation	L. K. Mansur, B. J. Frame, N. C. Gallego, J. W. Klett, L. W. Townsend, "Progress in Materials Science and Particle Transport Assessment of Novel and Multifunctional GCR Shielding Materials," Marshall Space Flight Center, Huntsville, Alabama, October 15, 2004 Oct-2004